

ABU SIMBEL, Egypt

WMO: 624190

Lat: 22.376N

Lon: 31.612E

Elev: 188

StdP: 99.09

Time Zone: 2.00 (E02)

Period: 06-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	8.0	9.2	-8.2	1.9	21.4	-6.2	2.3	20.0	9.5	18.1	8.7	18.3	2.7	340	0.478

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	14.2	44.0	21.5	43.0	21.1	42.0	20.8	23.6	38.2	22.7	38.4	22.1	39.0	4.5	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.4	13.6	34.1	16.8	12.2	33.9	15.2	11.0	32.8	71.3	38.1	67.9	38.9	65.3	38.8	26.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.8	8.0	7.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	27.0	16.7	19.3	22.9	27.5	31.4	34.0	33.8	34.8	32.6	29.3	23.2	18.0
	DBStd	7.00	3.09	3.43	3.84	3.45	2.78	2.09	1.77	1.65	2.02	3.32	3.05	2.70
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	143	73	25	4	0	0	0	0	0	0	0	1	39
	CDD10.0	6201	207	259	401	527	663	719	738	768	677	598	396	247
	CDD18.3	3302	21	51	147	277	405	469	479	510	427	339	148	28
	CDH23.3	51144	233	577	1781	3793	6396	8012	8245	8826	6812	4685	1531	255
	CDH26.7	33432	63	208	842	2198	4231	5687	5839	6379	4552	2781	606	46
Wind	WSAvg	3.6	3.4	3.6	3.7	3.5	3.7	3.7	3.0	3.5	3.9	3.7	3.5	3.6
Precipitation	PrecAvg	0	0	0	0	0	0	0	0	0	0	0	0	0
	PrecMax	4	4	0	0	0	0	0	0	0	0	0	0	0
	PrecMin	0	0	0	0	0	0	0	0	0	0	0	0	0
	PrecStd	1	1	0	0	0	0	0	0	0	0	0	0	0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.8	35.4	39.7	42.9	44.1	45.8	44.5	44.7	43.2	42.1	35.8	30.0
		MCWB	18.1	18.3	19.9	20.3	20.6	21.3	21.5	21.7	21.1	20.9	20.2	17.5
	2%	DB	28.1	31.8	36.8	40.1	42.6	44.0	43.1	43.5	41.8	40.0	33.2	28.0
		MCWB	15.8	16.6	18.5	19.2	20.4	21.2	21.0	21.9	21.0	20.8	19.3	16.2
	5%	DB	25.9	29.1	33.9	37.8	41.0	42.2	42.0	42.3	40.2	38.1	31.8	26.1
		MCWB	14.4	15.2	17.2	18.4	19.9	20.6	20.8	21.5	20.5	20.4	18.7	15.3
	10%	DB	23.8	27.0	31.1	35.9	39.2	41.1	40.9	41.2	39.2	36.4	30.1	24.3
		MCWB	13.4	14.4	15.6	17.5	19.1	20.1	20.4	21.2	20.3	20.2	17.6	14.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.8	19.4	20.9	21.2	22.9	23.3	24.7	25.0	23.2	23.1	21.8	18.7
		MCDB	29.5	32.3	36.0	39.2	39.4	40.1	38.8	38.8	38.9	36.8	32.3	27.0
	2%	WB	16.7	17.2	19.1	20.0	21.6	22.2	22.9	23.9	22.3	22.0	20.6	17.3
		MCDB	27.0	30.1	34.6	36.9	39.4	40.9	38.9	38.1	38.8	36.5	32.0	26.2
	5%	WB	15.0	15.8	17.6	19.0	20.6	21.4	21.7	23.0	21.5	21.3	19.4	16.1
		MCDB	25.1	28.2	32.5	36.0	38.7	40.6	39.7	37.7	38.4	36.0	30.3	24.9
	10%	WB	13.6	14.6	16.3	18.1	19.7	20.7	21.0	22.3	20.9	20.6	18.2	14.8
		MCDB	22.7	26.2	30.6	34.5	37.8	39.9	39.4	38.8	37.9	35.3	29.0	23.8
Mean Daily Temperature Range	5% DB	MDBR	13.2	13.7	15.0	14.8	14.9	14.7	14.7	14.2	14.0	13.3	13.3	12.9
		MCDBR	15.3	16.3	17.2	16.8	16.7	15.4	15.5	14.7	14.6	14.4	13.7	13.7
		MCWBR	8.1	7.9	7.4	6.7	6.9	6.2	6.1	5.7	5.9	6.1	7.1	7.3
	5% WB	MCDBR	15.1	15.9	16.6	16.2	15.4	14.5	14.4	13.1	13.8	13.7	13.1	12.9
		MCWBR	8.2	8.0	7.4	6.9	6.8	6.1	6.4	5.9	6.2	6.3	7.3	7.2
Clear-Sky Solar Irradiance	taub		0.379	0.422	0.464	0.553	0.575	0.527	0.548	0.578	0.540	0.499	0.407	0.383
	taud		2.152	2.006	1.880	1.694	1.666	1.790	1.773	1.729	1.792	1.871	2.083	2.134
	Ebn at Noon		877	860	840	773	751	781	765	744	764	775	839	856
	Edh at Noon		138	170	201	247	252	221	225	235	217	192	146	136
All-Sky Solar Radiation	RadAvg		5.08	5.93	6.97	7.67	7.91	8.06	7.83	7.39	6.87	6.04	5.25	4.79
	RadStd		0.14	0.17	0.17	0.17	0.17	0.11	0.13	0.14	0.14	0.08	0.09	0.07

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page